

One Tree, Four Distributions

A pedagogical route from Pascal's triangle to the probability of permutations, partitions, and ties

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Abstract

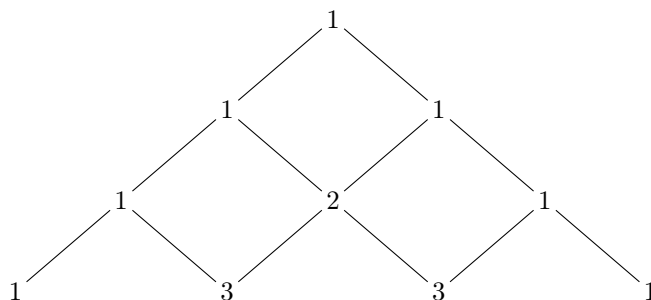
Probability is, for many pupils, the least secure topic in the syllabus, and the named distributions can feel arbitrary. This note builds a single picture that pupils already half-possess, Pascal's triangle as the count of paths down a branching tree, and shows that the binomial, the permutation-ascent distribution, the set-partition (Bell) numbers, and the two-sample rank statistic are all the same picture under a single modification: the branching rule of the tree is changed. Pascal is the tree with two equal branches. Changing the number of branches gives the Stirling and Bell numbers; changing the weights on the branches gives the Eulerian numbers and, with them, the distribution of a sum of uniform variables; adding a third branch gives the Delannoy numbers, and the third branch turns out to be a statistical tie. The teaching sequence is the incline; the destination is the recognition that one combinatorial idea organises four distributions a syllabus usually presents as unrelated.

1 The picture every pupil already has

Pascal's triangle is most securely taught as a branching path. Begin at a single root. At each step the path may go down-left or down-right. After n steps the path has made n choices. The picture that matters is the one in which paths arriving at the same point are allowed to *merge*: a path that has gone right k times out of n ends at the same lattice point as any other path that has gone right k times, whatever the order of its steps. The number written at a lattice point is the number of distinct paths that reach it, and that number is the entry of Pascal's triangle,

$$\binom{n}{k} = \binom{n-1}{k} + \binom{n-1}{k-1},$$

because a path reaching (n, k) arrived either from $(n-1, k)$ by a left step or from $(n-1, k-1)$ by a right step. The two incoming branches are added at the merge. *The recurrence is the merge.*



Reading the merge points by depth recovers the rows 1; 1, 1; 1, 2, 1; 1, 3, 3, 1. The interior 2 and the two interior 3s are exactly the points where paths converge, and their values are the sums of the two parents above. This distinction between a freely branching tree, whose leaves

would each be reached by a single path, and a merging lattice, whose interior counts grow, is worth making explicit: the binomial coefficients are large precisely because many distinct paths converge on the same point. If each branch is taken with probability one half, the probability of arriving at position k in row n is

$$\frac{1}{2^n} \binom{n}{k},$$

the binomial distribution with n trials and success probability one half. The Galton board, is this lattice made physical, and the heap of beads at the bottom is the binomial distribution drawn by gravity. This is the secure foothold. Everything that follows changes the tree in exactly one respect at a time.

2 First change: vary the number of branches

In Pascal's tree every node has two branches. Suppose instead that the number of branches depends on where the path is. This is not an abstract move; it is what happens when the tree counts the building-up of a set partition.

Build a partition of $\{1, 2, \dots, n\}$ one element at a time. Element 1 starts a single block. When element m arrives, with the first $m - 1$ elements already arranged into some number of blocks, m has a choice: it may join any one of the existing blocks, or it may open a new block of its own. If there are currently k blocks, the new element faces $k + 1$ choices, k of them joining and one of them opening. The number of partitions of m things into k blocks therefore obeys

$$S(m, k) = k S(m - 1, k) + S(m - 1, k - 1),$$

the Stirling recurrence of the second kind. Compare it with Pascal's. The second term is identical, the opening of a new block playing the role of the right step. The first term carries a weight k where Pascal carried 1: a node sitting at k blocks has k branches that lead back to k blocks, not one. Pascal's tree is the special case in which that weight is frozen at unity.

The total number of partitions of an n -set, summing over all block counts, is the Bell number

$$B_n = \sum_k S(n, k), \quad B_0, B_1, \dots = 1, 1, 2, 5, 15, 52, 203, \dots$$

and the row sums of the varying-branch tree are these Bell numbers, where the row sums of Pascal's two-branch tree were the powers of two. A pupil who can see that "two branches always" gives 2^n can see that " $k + 1$ branches at a k -block node" gives something larger and less regular, and that the something is the count of all set partitions.

2.1 Where the probability lives

The probabilistic home of the Bell numbers is the Poisson distribution, and the link is worth stating even if its proof is beyond the pupil. The n th moment of a Poisson random variable with mean one is exactly B_n ; equivalently, by Dobinski's formula,

$$B_n = \frac{1}{e} \sum_{k \geq 0} \frac{k^n}{k!}.$$

The integers that count set partitions are the moments that describe the spread of the most basic counting process in probability. The headline is enough: the same numbers that count "ways to sort people into unlabelled groups" measure "how spread out a random count is."

3 Second change: vary the weights on the branches

Return to two branches, but now let the two branches carry different weights, and let those weights depend on the state. This is what happens when the tree counts permutations by their number of ascents.

Build a permutation of $\{1, 2, \dots, n\}$ by inserting the elements in order. Suppose a permutation of the first $n - 1$ elements has k ascents, that is, k places where a value is followed by a larger one. Insert the element n , the largest so far, into one of the n available gaps. Inserting it into a gap that already sits at an ascent, or at the right-hand end, leaves the ascent count unchanged or raises it in a controlled way; inserting it elsewhere changes the count differently. Counting the gaps of each type gives the recurrence

$$\langle n, k \rangle = (k + 1) \langle n - 1, k \rangle + (n - k) \langle n - 1, k - 1 \rangle,$$

where $\langle n, k \rangle$ is the Eulerian number, the count of permutations of n elements with exactly k ascents. Once more the shape is Pascal's, two terms feeding each node, but now *both* branches carry weights, $k + 1$ on one and $n - k$ on the other, and both depend on position. Pascal is recovered if both weights are set to one, which they never are here.

The rows are

$$1; \quad 1, 1; \quad 1, 4, 1; \quad 1, 11, 11, 1; \quad 1, 26, 66, 26, 1,$$

and they sum to $n!$, since every one of the $n!$ permutations has some number of ascents. The two-branch tree with unit weights counted 2^n outcomes; the two-branch tree with these position-dependent weights counts $n!$ outcomes.

3.1 Where the probability lives

Dividing a row by $n!$ gives the exact distribution of the ascent count of a random permutation. For $n = 4$ the distribution is

$$\frac{1}{24}(1, 11, 11, 1),$$

the probabilities that a random shuffle of four items has 0, 1, 2, 3 ascents. This is already a genuine, classroom-accessible distribution: shuffle four cards, count the rises, and these are the odds.

The same numbers reach into continuous probability. If $U_1, \dots, U_n$ are independent and uniform on $[0, 1]$ and $S = U_1 + \dots + U_n$ is their sum, then the probability that S falls in the interval $[k, k + 1)$ is exactly $\langle n, k \rangle / n!$. For $n = 3$ these probabilities are $\frac{1}{6}, \frac{2}{3}, \frac{1}{6}$; for $n = 4$ they are $\frac{1}{24}, \frac{11}{24}, \frac{11}{24}, \frac{1}{24}$. The Eulerian numbers, which count ascents, are also the integer-interval masses of the density of a sum of uniform variables. That density, the Irwin–Hall or uniform-sum distribution, is the continuous shadow of the same tree, and as n grows it approaches the bell-shaped normal curve, which is the central limit theorem appearing inside the very same construction that began with Pascal.

4 Third change: add a branch

The deepest change is also the simplest to state. Give every node a third branch.

Pascal's tree had two branches, left and right. Add a third, a diagonal step that moves the path forward in both coordinates at once. The count of paths to a node now obeys

$$D(n, k) = D(n - 1, k) + D(n, k - 1) + D(n - 1, k - 1),$$

with three predecessors rather than two, and the resulting numbers are the Delannoy numbers. The third term, the diagonal, is the new branch. Because of it the array is most naturally drawn not as a triangle but as a square, and the central entries 1, 3, 13, 63, 321, 1683 grow far faster than the binomial entries of the two-branch tree.

4.1 Where the probability lives, and why it is different

Here the pattern of the previous sections breaks, and the break is the most instructive moment in the whole sequence. The first three trees each gave a distribution along a single index: a number of heads, a number of ascents, a number of blocks. The three-branch tree does not. To see why, consider the two-sample rank test that pupils meet, if they meet it at all, late and uneasily: two groups of measurements are pooled and ranked, and one asks whether the two groups are intermingled or separated.

Walk through the pooled ranking from smallest to largest. Step east when the next value comes from the first group and north when it comes from the second. Every way the two groups can interleave is a path using east and north steps, and the number of such paths is a binomial coefficient: the two-branch tree again, now counting interleavings. So far the rank test is Pascal in disguise.

Then two measurements, one from each group, turn out to be equal. They are tied. In the ranking neither comes before the other; they advance together. That is a single diagonal step, the third branch, consuming one unit from each group at once. The moment ties are allowed, the two-branch interleaving count becomes the three-branch Delannoy count. For two groups of two the tie-free count is $\binom{4}{2} = 6$ and the count allowing ties is the Delannoy number 13; for groups of three the counts are 20 and 63.

The third branch is the tie.

This is why the three-branch tree gives no single-index distribution. A distribution along one index requires that outcomes be totally ordered, each comparable to every other: more heads or fewer, more ascents or fewer. A tie is exactly the situation in which two things cannot be ordered, because they are equal. Ties are a partial order, not a total one, and a partial order does not collapse onto a single line. It needs a second dimension. The third branch of the tree, the diagonal, is the geometric form of that second dimension, and it is why the Delannoy numbers live in a square array and govern a two-sample, two-dimensional comparison rather than a one-dimensional distribution.

5 The sequence, seen whole

The teaching runs along a single incline one modification of the tree at a time.

Tree	Modification	Counts	Distribution
Two branches, equal	none (the base case)	$\binom{n}{k}$	Binomial
Branch <i>count</i> varies	$k + 1$ branches at k blocks	Stirling, Bell	Poisson moments
Branch <i>weights</i> vary	weights $k + 1, n - k$	Eulerian	ascents; uniform sum
A third branch added	diagonal step	Delannoy	two-sample test with ties

The binomial is the tree with two equal branches. Vary how many branches leave each node and the partitions of a set appear, with the Poisson distribution behind them. Vary the weights the branches carry and the permutations appear, sorted by their ascents, with the distribution of a sum of uniform numbers behind them. Add a branch and the tie appears, and with it the two-sample rank test, which is not a single distribution at all but a two-dimensional comparison, because a tie is the one thing a single ordering cannot express.

A pupil who holds the bifurcating tree securely can be walked up this incline one step at a time, and at the top will have met four of the distributions a syllabus usually presents as unrelated arrivals, and will have seen that they are one construction seen under four small changes of a single rule. The hardest topic becomes, for once, a series of variations on a picture already owned.

A The worked recurrences at a glance

Each distribution is the same node-count recurrence with a different rule. In every case a path reaches a node from its predecessors; only the number of predecessors and their weights change.

Binomial	$\binom{n}{k} = \binom{n-1}{k} + \binom{n-1}{k-1}$
Stirling (2nd kind)	$S(n, k) = k S(n-1, k) + S(n-1, k-1)$
Eulerian	$\langle n, k \rangle = (k+1) \langle n-1, k \rangle + (n-k) \langle n-1, k-1 \rangle$
Delannoy	$D(n, k) = D(n-1, k) + D(n, k-1) + D(n-1, k-1)$

One worked node in each.

- Binomial: $\binom{4}{2} = \binom{3}{2} + \binom{3}{1} = 3 + 3 = 6$.
- Stirling: $S(4, 2) = 2 S(3, 2) + S(3, 1) = 2 \cdot 3 + 1 = 7$.
- Eulerian: $\langle 4, 1 \rangle = 2 \langle 3, 1 \rangle + 3 \langle 3, 0 \rangle = 2 \cdot 4 + 3 \cdot 1 = 11$.
- Delannoy: $D(2, 2) = D(1, 2) + D(2, 1) + D(1, 1) = 5 + 5 + 3 = 13$.

Sources

The identity of the Eulerian numbers with the integer-interval masses of a sum of uniform variables is due to S. Tanny, “A probabilistic interpretation of Eulerian numbers,” *Duke Mathematical Journal* 40 (1973), 717–722. The continuous distribution so obtained is the Irwin–Hall (uniform sum) distribution. Dobinski’s formula expresses the Bell numbers as the moments of a unit-mean Poisson variable. The lattice-path reading of the Wilcoxon–Mann–Whitney two-sample statistic, in which tied observations correspond to diagonal steps and the tied-case path count becomes the Delannoy numbers, follows the combinatorial treatment of that statistic’s null distribution in the statistical literature. All numerical values in this note were checked directly.

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